

AY 25-26 SI

Quiz 1 Solutions

1. $3x + 6y - 5z = -3$

$$-2x + 7y - z = 24.$$

$$\begin{bmatrix} 3 & 6 & -5 & -3 \\ -2 & 7 & -1 & 24 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + \frac{2}{3}R_1} \begin{bmatrix} 3 & 6 & -5 & -3 \\ 0 & 11 & -\frac{13}{3} & 22 \end{bmatrix}$$

$$\rightarrow 3x + 6y - 5z = -3$$

$$11y - \frac{13}{3}z = 22$$

$$\Rightarrow y = \frac{1}{11} \left(\frac{13}{3}z + 22 \right) = 2 + \frac{13}{33}z$$

$$\rightarrow 3x + 6 \left(2 + \frac{13}{33}z \right) - 5z = -3$$

$$\Rightarrow x = -5 + \frac{29}{33}z$$

z is a free variable; let $z = t$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 29/33 \\ 13/33 \\ 1 \end{bmatrix}$$

$$\Rightarrow \vec{r} = \begin{bmatrix} -5 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 29 \\ 13 \\ 33 \end{bmatrix}.$$

B

2. Let $\vec{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

$$\left. \begin{array}{l} 2a + 2b + c = 0 \\ 2a - 2b + 3c = 0 \end{array} \right\} \rightarrow \begin{array}{l} 4b - 2c = 0 \\ \Rightarrow b = \frac{1}{2}c \end{array}$$

$$\begin{aligned} a^2 + b^2 + c^2 &= 3^2 = 9 \\ \Rightarrow c^2 + \frac{1}{4}c^2 + c^2 &= 9 \end{aligned}$$

$$\Rightarrow \frac{9}{4}c^2 = 9 \Rightarrow c = \pm 2$$

$$b = \pm 1, \quad a = -2$$

$$\therefore \vec{x} = \begin{bmatrix} -2 \\ \pm 1 \\ \pm 2 \end{bmatrix} = \pm (2\vec{i} \mp \vec{j} \mp 2\vec{k})$$

$$= \pm (2\vec{i} - \vec{j} - 2\vec{k}) \quad \boxed{A}$$

$$3. \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{array}{l} 2b_{11} + b_{13} = 1 \quad b_{11} = 1 \\ b_{11} + b_{12} = 0 \Rightarrow b_{12} = -1 \\ b_{11} + b_{13} = 0 \quad b_{13} = -1 \end{array}$$

$$\begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \Rightarrow \begin{array}{l} 2b_{21} + b_{23} = 2 \quad b_{21} = 2 \\ b_{21} + b_{22} = 3 \Rightarrow b_{22} = 1 \\ b_{21} + b_{23} = 0 \quad b_{23} = -2 \end{array}$$

$$\begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} b_{31} \\ b_{32} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \begin{array}{l} 2b_{31} + b_{33} = 3 \\ b_{31} + b_{32} = 2 \\ b_{31} + b_{33} = 1 \end{array}$$

$$\begin{aligned} \therefore b_{11} + b_{22} + b_{33} \\ = 1 + 1 - 1 = 1 \end{aligned}$$

$$\Rightarrow \begin{array}{l} b_{31} = 2 \\ b_{32} = 0 \\ b_{33} = -1 \end{array}$$

OR

$$AB = [Ab_1 \quad Ab_2 \quad Ab_3] = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix} = C$$

$$\Rightarrow B = A^{-1}C = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

D

$$\Rightarrow 1 + 1 - 1 = 0.$$

4. $AB = A$, $BA = B$

$$BAB = B^2 \Rightarrow B(AB) = B^2 \Rightarrow BA = B^2 \Rightarrow B = B^2$$

OR $AB = A \Rightarrow B = I$
 $BA = B \Rightarrow A = I$

$$\therefore B^2 = A^2, \quad B^2 = A$$

If you have marked (A) or (B) or (C)
or any combination of them, you
get full marks.

5. The set $\{\vec{v}, a\vec{v}\}$ is linearly dependent, but its subset $\{\vec{v}\}$ is linearly independent.

False

6. Since A is 4×3 , there is no pivot in every row. According to Theorem 1.2 of lectures, there is no solution for some $\vec{b}$.

B

7. $A\vec{x} = \vec{0}$ has infinite solutions means A is singular (i.e. no inverse).

$$\Rightarrow \begin{vmatrix} k & 2k \\ k^2 - k & k^2 \end{vmatrix} = 0 \Rightarrow k^3 - 2k(k^2 - k) = 0$$

$$\Rightarrow k^3 - 2k^3 + 2k^2 = 0$$

$$\Rightarrow k^3 - 2k^2 = 0$$

$$\Rightarrow k^2(k - 2) = 0$$

$$\Rightarrow k = 0 \text{ or } 2.$$

B

8. True:

e.g. for $\mathbb{R}^2$, $\text{span}\{\vec{u}, \vec{v}\} = \mathbb{R}^2$

and $\text{span}\{\vec{u}, \vec{u} + \vec{v}\}$ is also $\mathbb{R}^2$.

9. For a basis of null space of T ,
the sum of the diagonal elements,
i.e. the trace of the matrix
should be 0.

$\therefore \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ is not a basis.

C

10. D

11. 3 vectors cannot span five dimensional
space. Those vectors can be linearly
independent or dependent.

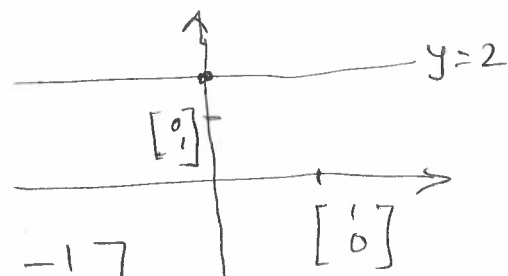
C

$$12. A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix}$$

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 0 & -1 \\ 2 & 2 \end{bmatrix}$$



E

13. True

$t \begin{bmatrix} 3 \\ 0 \\ -7 \end{bmatrix}$ is a subspace of $\mathbb{R}^3$.

$$14. \quad A^2 = \begin{bmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{bmatrix} \begin{bmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{bmatrix} = \begin{bmatrix} \sin^2 \alpha & 0 \\ 0 & \sin^2 \alpha \end{bmatrix}$$

$$|A^2 - \frac{1}{2}I| = \begin{vmatrix} \sin^2 \alpha - \frac{1}{2} & 0 \\ 0 & \sin^2 \alpha - \frac{1}{2} \end{vmatrix}$$

$$= \left(\sin^2 \alpha - \frac{1}{2}\right)^2 = \sin^4 \alpha - \sin^2 \alpha + \frac{1}{4} \neq 0.$$

$$\Rightarrow \sin^2 \alpha = \frac{1}{2} \Rightarrow \sin \alpha = \frac{1}{\sqrt{2}} \Rightarrow \alpha = \frac{\pi}{4}.$$

C

$$15. \quad A = \begin{bmatrix} \textcircled{2} & 4 & 1 & 3 \\ 6 & 13 & 5 & 7 \\ -2 & -3 & 0 & 1 \end{bmatrix} \begin{array}{l} R_2 \leftarrow R_2 - 3R_1 \\ \longrightarrow \\ R_3 \leftarrow R_3 + R_1 \end{array}$$

$$A' = \begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & \textcircled{1} & 2 & -2 \\ 0 & 1 & 1 & 4 \end{bmatrix} \begin{array}{l} R_3 \leftarrow R_3 - R_2 \\ \longrightarrow \end{array}$$

$$\begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & -1 & 6 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ ? & 1 & 0 \\ ? & ? & 1 \end{bmatrix}$$

Divide 1st col. of A by 2 :

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -1 & ? & 0 \end{bmatrix}$$

Divide 2nd col of A' by 1 :

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

B

16.
$$\begin{bmatrix} 0 & -1 & -1 & \dots & -1 \\ -1 & 0 & & & \\ \vdots & & & & \\ -1 & & & & 0 \end{bmatrix}$$

Add all rows to the last row :

$$\begin{bmatrix} 0 & -1 & -1 & \dots & -1 \\ \vdots & & & & \\ -(n-1) & -(n-1) & \dots & -(n-1) \end{bmatrix}$$

Take $-(n-1)$ out

$$(1-n) \begin{vmatrix} 0 & -1 & -1 & \dots & -1 \\ -1 & 0 & -1 & \dots & -1 \\ -1 & -1 & 0 & \dots & -1 \\ \vdots & & & & \\ 1 & 1 & 1 & \dots & 1 \end{vmatrix}$$

Add last row back to each row

$$(1-n) \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & & & \ddots & \\ 0 & & & & 1 \end{vmatrix} = (1-n)$$

~~is~~ diagonal

D

$$17. \begin{bmatrix} 6 & -4 \\ -3 & 2 \\ -9 & 6 \\ 9 & -6 \end{bmatrix} \xrightarrow{\substack{R_2 \leftarrow R_2 + \frac{1}{2} R_1 \\ R_3 \leftarrow R_3 + \frac{3}{2} R_1}} \begin{bmatrix} 6 & -4 \\ 0 & 0 \\ 0 & 0 \\ 9 & -6 \end{bmatrix}$$

$$\xrightarrow{R_4 \leftrightarrow R_2} \begin{bmatrix} 6 & -4 \\ 9 & -6 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - \frac{3}{2} R_1} \begin{bmatrix} 6 & -4 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$6x - 4y = 0 \Rightarrow 6x = 4y$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = t \begin{bmatrix} 2/3 \\ 1 \end{bmatrix}$$

subspace of $\mathbb{R}^2$

B

18.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x &= 0 \\ y &= 0 \end{aligned}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\underbrace{\hspace{1cm}}_{z\text{-axis}}$

C

19.

$$\begin{bmatrix} 1 & 2 & b_1 \\ 2 & 4 & b_2 \\ 3 & 7 & b_3 \\ 3 & 9 & b_4 \end{bmatrix} \begin{array}{l} R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - 3R_1 \\ R_4 \leftarrow R_4 - 3R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & b_1 \\ 0 & 0 & b_2 - 2b_1 \\ 0 & 1 & b_3 - 3b_1 \\ 0 & 3 & b_4 - 3b_1 \end{bmatrix} \begin{array}{l} R_2 \leftrightarrow R_3 \\ \longrightarrow \end{array}$$

$$\begin{bmatrix} 1 & 2 & b_1 \\ 0 & 1 & b_3 - 3b_1 \\ 0 & 0 & b_2 - 2b_1 \\ 0 & 3 & b_4 - 3b_1 \end{bmatrix} \begin{array}{l} R_4 \leftarrow R_4 - 3R_2 \\ \longrightarrow \end{array}$$

$$\begin{bmatrix} 1 & 2 & b_1 \\ 0 & 1 & b_3 - 3b_1 \\ 0 & 0 & b_2 - 2b_1 \\ 0 & 0 & b_4 + 6b_1 - 3b_3 \end{bmatrix}$$

For consistent equations,

$$b_2 - 2b_1 = 0 \quad \& \quad b_4 + 6b_1 - 3b_3 = 0$$

A

20. Correct answer not in the choices.

Everyone gets ~~5~~ full marks for this question.